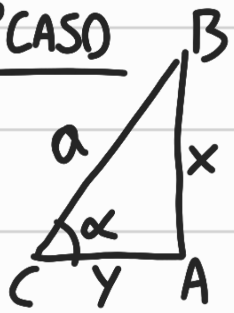
$1^{\underline{o}}$ CASO: $\sqrt{a^2-x^2}$ (INTEGRANDO)

Considere o triângulo retângulo

$1^{\underline{o}}$ CASO



$0 < \alpha < 90°$

CATETO ADJACENTE

COTANGENTE

I) $x^2+y^2=a^2$

$y=\sqrt[2]{x^2-a^2}$

II) $\operatorname{sen}\alpha = \frac{x}{a}$

$x = a \cdot \operatorname{sen}\alpha$

$y = a.\cos\alpha$

III) $\operatorname{cotg}\alpha = \frac{\cos\alpha}{\operatorname{sen}\alpha}$

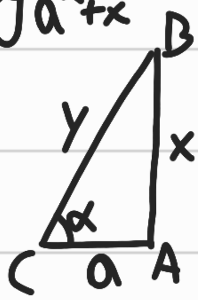
$\operatorname{cotg}\alpha = \frac{y}{x}$ $\quad \operatorname{cotg}\alpha = \frac{\sqrt[2]{a^2-x^2}}{x}$

$x = a.\operatorname{sen}x \Rightarrow dx = a.\cos\alpha$

$2^{\underline{o}}$ CASO: HIPOTENUSA

$\sqrt[2]{a^2+x^2}$

SECANTE



I) $x = y \cdot \operatorname{sen}x$ $\quad y = a \cdot \sec x$

OU

$x = a \cdot \operatorname{tg}x$

II) $\sec\alpha = \frac{\sqrt{a^2+x^2}}{x}$

$x = a\operatorname{tg}\alpha \Rightarrow dx = a.\sec^2\alpha\, dx$

$3^{\underline{o}}$ CASO. $\sqrt[2]{x^2-a^2}$

$y = a.\operatorname{tg}\alpha$

$x = a.\sec\alpha$

CATETO OPOSTO

$x = a.\sec\alpha$

$dx = a \cdot \sec\alpha \cdot \operatorname{tg}\alpha\, dx$